Ayush Chauhan

Pre-Final Year Graduate, VIT Vellore

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PROFILE

Computer Science undergraduate at VIT Vellore (CGPA: 8.64) with proven experience building and shipping end-to-end ML and full-stack systems. Designed a deployed stock intelligence platform integrating Gradient Boosting, NLP sentiment analysis, and Claude Vision API; built a real-time facial recognition pipeline achieving 88% accuracy at 20 FPS. Proficient across the full development lifecycle — data engineering, model training, REST APIs, and production deployment on AWS and Streamlit.

EDUCATION

B.Tech, Computer Science Engineering — CGPA: 8.64 Aug 2023 – May 2027

Vellore Institute of Technology — Vellore, India

- Relevant Coursework: Machine Learning, Data Structures & Algorithms, Database Management Systems, Operating Systems, Object-Oriented Programming

PROFESSIONAL EXPERIENCE

Software Development Intern Jun 2024 – Jul 2024

Rostan Technologies — Gurugram, India

- Developed 5+ data-facing Oracle APEX modules for an internal Scanner Application supporting 50+ daily document scans, improving team workflow efficiency by 20%.
- Optimized backend data retrieval pipelines by refactoring PL/SQL queries and introducing indexed joins, reducing average latency by 15%.
- Enhanced mobile responsiveness and UI usability across APEX pages, reducing user-reported friction by 30%.
- Collaborated with cross-functional stakeholders to gather requirements, translating business workflows into modular application components and reducing manual data-entry errors by ~25%.
- Implemented input validation and error-handling routines for document scanning forms, improving data integrity and reducing erroneous entries in production by over 40%.
- Documented module architecture and API interfaces, enabling faster onboarding for two subsequent interns and reducing knowledge-transfer time by an estimated 50%.

PROJECTS

NiftyMind — AI-Powered Indian Stock Intelligence Platform [Live App]

Python · scikit-learn · Streamlit · VADER NLP · Claude Vision API

- Architected a full-stack AI stock intelligence platform for NSE/BSE markets integrating ML-based price prediction, NLP sentiment analysis, and AI-powered chart analysis via Claude Vision API into a single deployed Streamlit application.
- Engineered a Gradient Boosting Regressor for next-day and 5-day stock price forecasting using a 65-feature matrix built from a 60-day rolling window with recursive multi-step prediction and cross-validation confidence intervals.
- Implemented a 3-model parallel IPO success predictor (listing gain, 30-day return, subscription label) trained on 58 NSE IPOs spanning 2015–2025, using GMP, institutional subscription multiples, and NIFTY trend as features.
- Constructed a portfolio optimizer using Markowitz Mean-Variance theory with SLSQP constrained optimization and a 2,000-point Monte Carlo efficient frontier simulation supporting 3 risk strategies across up to 10 stocks.
- Designed a 6-layer data waterfall (NSE India → Alpha Vantage → Stooq → yfinance → CSV fallback) ensuring zero blank-chart failures under API rate limiting or source outages.

Real-Time Facial Recognition Attendance System [GitHub]

Python · OpenCV · LBPH Classifier · NumPy · Pandas

- Designed an end-to-end ML pipeline spanning data capture, preprocessing, feature extraction, and classification on live video streams at 20 FPS using OpenCV.
- Trained an LBPH classifier on 40+ user samples achieving 88% recognition accuracy; reduced attendance recording time by 90% — from 10 minutes to under 1 minute per session.
- Maintained 200+ timestamped attendance records in structured CSV format for downstream reporting and analysis.

TECHNICAL SKILLS

ML & Data:	Python, NumPy, Pandas, scikit-learn, OpenCV, Matplotlib, VADER NLP
Languages:	Java, C++, C
Web & Backend:	ReactJS, Angular, NodeJS, HTML, CSS, Oracle APEX
Databases:	SQL, MySQL, PL/SQL, MongoDB, Supabase
Tools & Platforms:	Git, Docker, AWS, Streamlit
Core CS:	Data Structures & Algorithms, OOP, DBMS, Operating Systems